Transformations of y=x

1

a Yes **b** No

2

a They become very large in the negative direction.

b They become very large in the positive direction.

3

a Increasing **b** Equally steep

4

a No **b** No **c** No **d** Yes

5

a No **b** No **c** No **d** Yes

6

a The constant is the same.

b All of the graphs cross the y-axis at the same point.

c Equations of the form $y = mx + b$ that have the same value of b generate graphs that cross the y-axis at the same point.

7

a $y = 1$ **b** b

8

a

i $x = 0$ **ii** $y = 0$

b

i $x = 3$ **ii** $y = 4$

c

i There is no x-intercept.

ii $y = -1$

9

a No **b** No **c** Yes **d** Yes

e No **f** Yes

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10 Positive

11

a Yes **b** Yes **c** No **d** No

e Yes

12

a $y = 5$ **b** $y = 0$ **c** $y = 10$ **d** $y = 1$

13

a Yes **b** No **c** Yes **d** Yes

e Yes

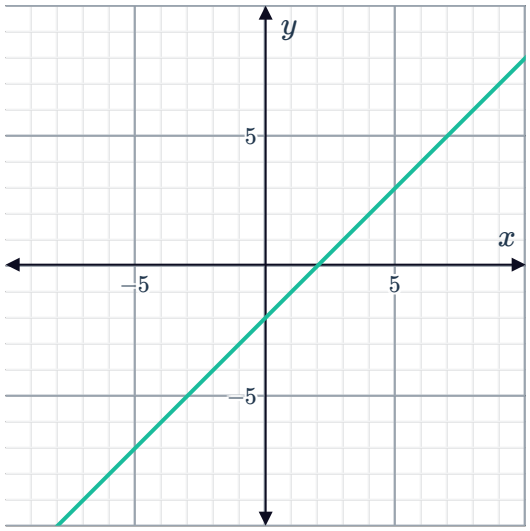
14

a $(4, 0)$ **b** $(0, -4)$ **c** $(0, -10)$

15

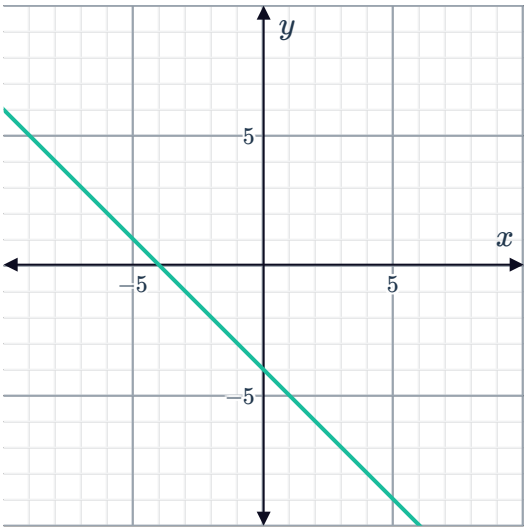
a $(4, 0)$ **b** $(0, 4)$ **c** $(0, 14)$

16

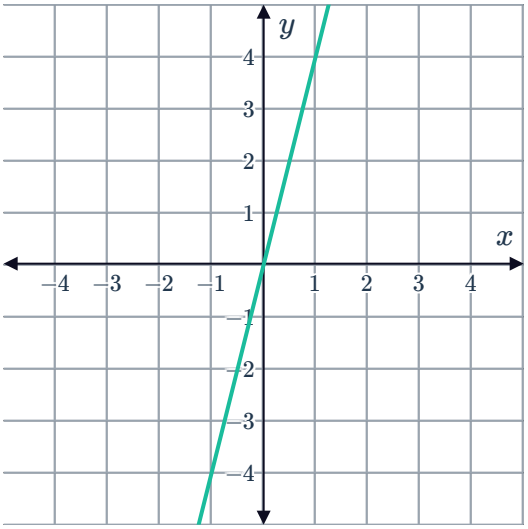


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17



18



19

a Yes **b** Yes **c** No **d** No

e No

20

a Yes **b** No **c** Yes **d** No

e No **f** No

21 $(1, 3)$

22

a Increasing

b Flatter

c

i No **ii** No **iii** No **iv** Yes

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23

a

i Yes **ii** No **iii** No **iv** Yes

v Yes

b

i No **ii** Yes **iii** Yes

c $x = 8$

24

a Decreasing

b Steeper

c

i Yes **ii** Yes **iii** No **iv** No

v No

25

a Statement 2: The set of points lie on a horizontal line.

b Statement 2: The set of points lie on a horizontal line.

c Statement 1: The set of points lie on a vertical line.

d Statement 3: The set of points lie on an increasing line.

e Statement 1: The set of points lie on a vertical line.

f Statement 4: The set of points lie on a decreasing line.

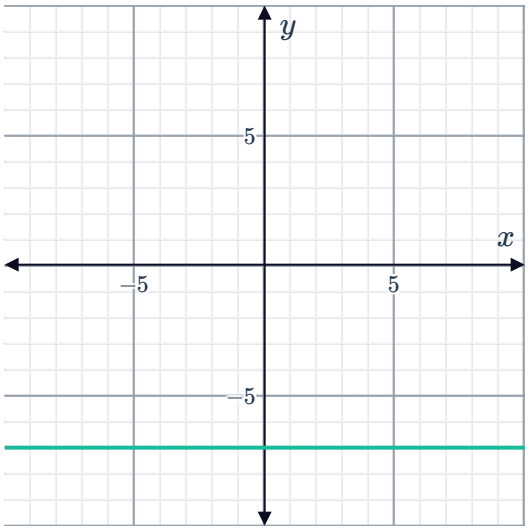
26

a A horizontal line **b** A vertical line

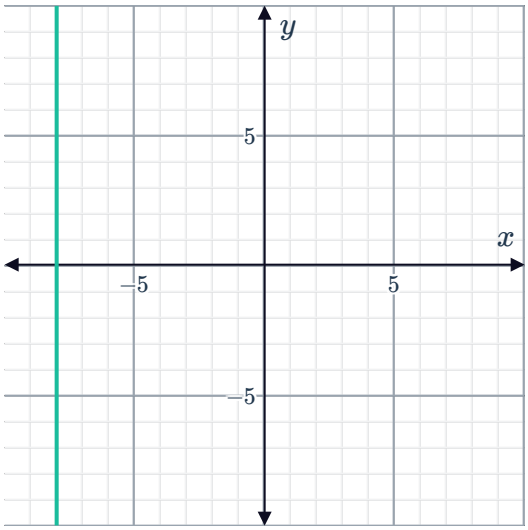
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27

a



b



28

a $x = -8$

b None

29

a $y = -4$

b None

30

a $x = -3$

b $y = 4$

31 $y = -4.5$

32 x-axis

33 y-axis

34

a $y = 13,\ y = -7$

b $x = 5,\ x = -9$

35 $y = 0,\ y = -4$

36

a $\$2.50$

b A fixed cost such as a toll charge

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Additional questions

37

a The transformation from $f(x)$ to $g(x)$ is a vertical translation.

b The transformation from $f(x)$ to $g(x)$ is making the line steeper.

c The transformation from $f(x)$ to $g(x)$ is a reflection across the x-axis.

38 When k is negative, the transformation will involve a reflection across the x-axis.

39

a Vertical translation

b Making the line steeper

c Making the line less steep

d Making the line less steep

e Reflection

f Vertical translation

40

a Vertical translation

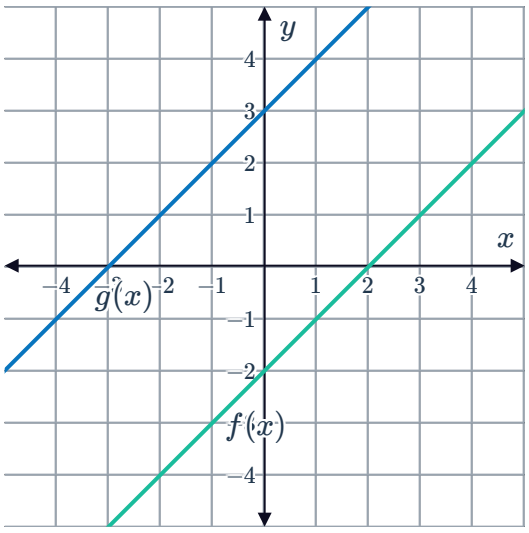
b Making the line steeper

c Making the line less steep and reflection across the x-axis

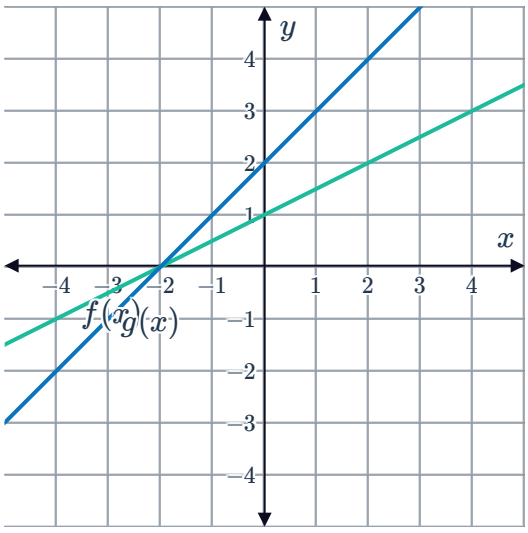
d Vertical translation

41

a

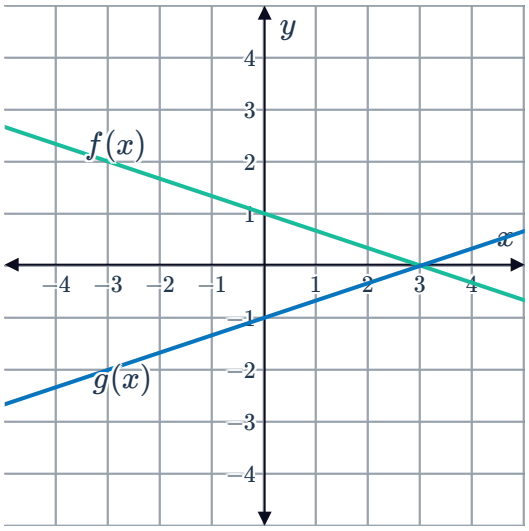


b

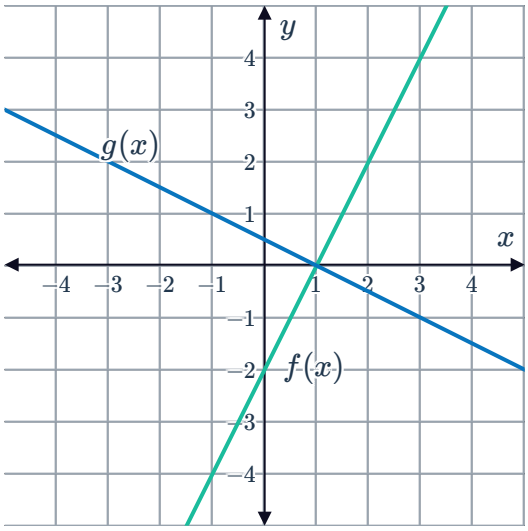


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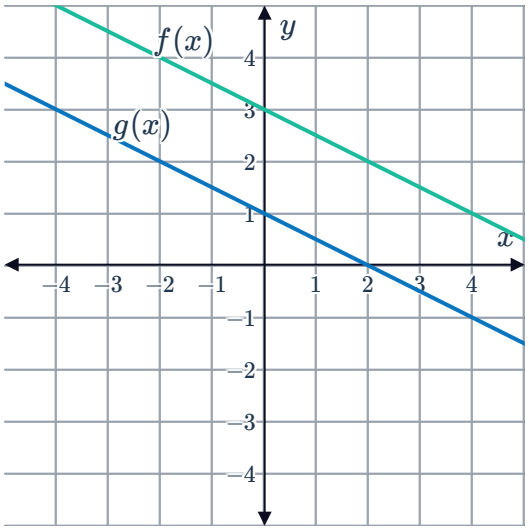
c



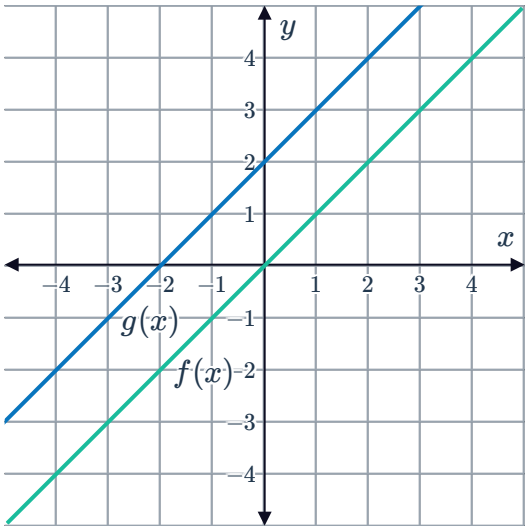
d



e



f



42

a The graph of $g(x)$ is the vertical translation of 8 units downwards from the graph of $f(x)$.

b The graph of $g(x)$ is the vertical translation of $\frac{1}{2}$ units upwards from the graph of $f(x)$.

c The graph of $g(x)$ is steeper than the graph of $f(x)$ by a factor of -3 and a reflection across the x-axis.

d The graph of $g(x)$ is making the graph of $f(x)$ steeper by a factor of $\frac{1}{4}$. This is equivalent to making the line less steep by a factor of 4.

43

a Oreste reflected the graph of $f(x)$ across the y-axis instead of the x-axis.

b If $f(x)$ is a linear function passing through the origin, then a reflection across the y-axis has the same effect as a reflection across the x-axis. If this is the case, Oreste's error would still give the correct answer.

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